

PCD Assignment 2
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1. Introduction and Objective

Real-world images are frequently degraded by non-ideal acquisition conditions: insufficient lighting produces dark, low-key images; excessive lighting or sensor overexposure produces washed-out, bright images; a narrow dynamic range produces flat, low-contrast images; and camera shake or defocus produces blurred images. This assignment implements and evaluates a targeted enhancement technique for each of these four degradation types. Because each degradation has a distinct underlying cause – exposure, dynamic range, and spatial-frequency loss – a single generic filter is not appropriate for all of them; instead, a purpose-fit classical image-processing technique is selected and applied to each case, and the outcome is evaluated both visually and with quantitative metrics.

2. Methodology

To allow a controlled, like-for-like comparison, four degraded test images were synthesised from a single clean base photograph: a **dark** image (pixel intensities scaled to 30% of the original), a **bright** image (intensities shifted upward by a constant offset of 140), a **low-contrast** image (dynamic range compressed into a narrow mid-gray band), and a **blurred** image (Gaussian blur with $\sigma = 6$). The same pipeline works identically on real photographs – only the image-loading step changes. Three quantitative metrics were used to evaluate every result: **mean intensity** (overall brightness), **intensity standard deviation** (a proxy for contrast), and **Laplacian variance** (a standard proxy for edge sharpness/focus, since blur suppresses high-frequency content and therefore lowers this value).

3. Dark (Underexposed) Image – Gamma Correction

A dark image has most pixel intensities compressed near zero. Gamma correction applies the power-law transform output = $255 \times (\text{input}/255)^{(1/\gamma)}$; using $\gamma = 2.6$ ($\gamma > 1$, so the exponent $1/\gamma < 1$) non-linearly stretches the shadow tones upward while compressing the already-bright pixels, revealing detail hidden in the dark regions without clipping highlights. Being a smooth curve rather than a hard stretch, it avoids introducing banding artifacts.

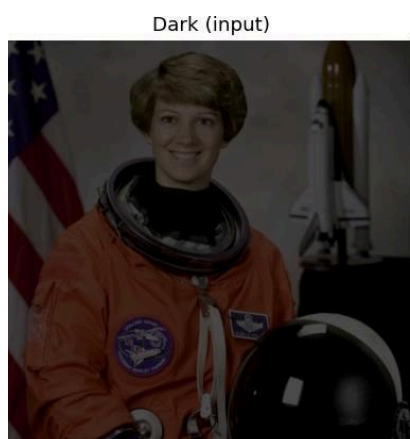


Figure 1. Dark input (left) vs. gamma-corrected output, $\gamma=2.6$ (right).

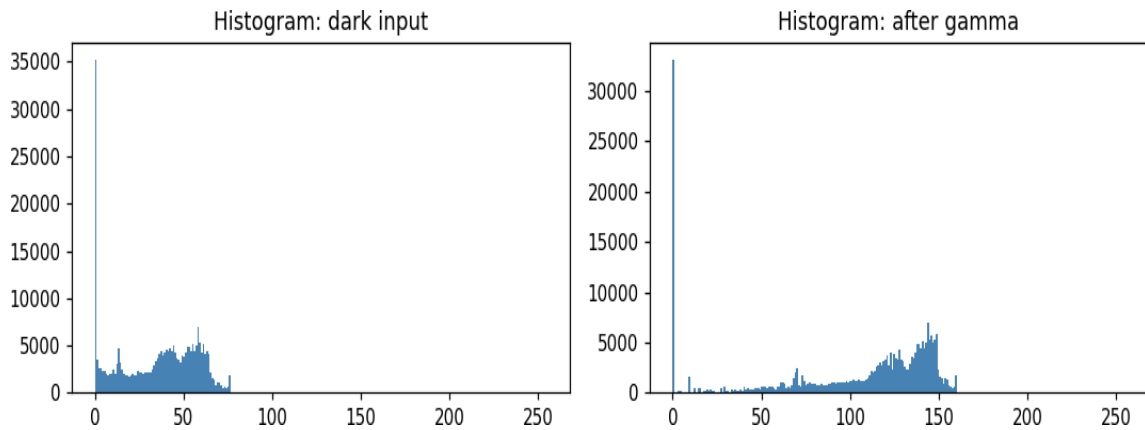


Figure 2. Histogram before (left, crushed near 0) and after (right, spread across mid-tones) gamma correction.

4. Bright (Overexposed) Image – Inverse Gamma Correction

An overexposed image suffers the mirror-image problem: intensities are pushed toward 255. The same gamma transform with $\gamma = 0.45$ ($\gamma < 1$, so the exponent $1/\gamma > 1$) pulls bright tones back down and expands separation in the highlight region, recovering detail that was nearly clipped to white. This comes at the cost of a modest reduction in overall brightness, an acceptable trade-off for restoring highlight detail.



Figure 3. Bright/overexposed input (left) vs. gamma-corrected output, $\gamma=0.45$ (right).

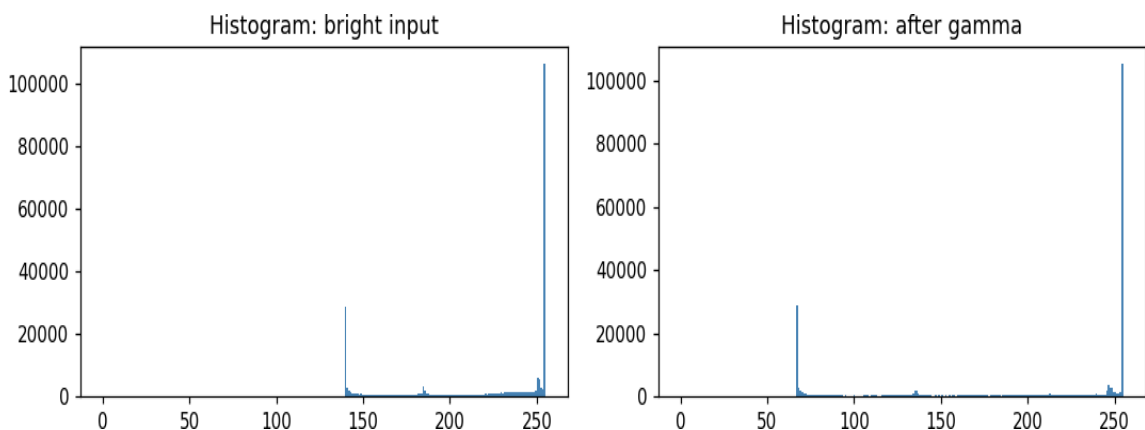


Figure 4. Histogram before (left, clipped near 255) and after (right, recovered highlight separation).

5. Low-Contrast Image – CLAHE

A low-contrast image has its histogram squeezed into a narrow band. Plain global histogram equalization was considered but rejected, since it tends to over-amplify noise and can distort regions that were already well-exposed. Instead, **CLAHE (Contrast Limited Adaptive Histogram Equalization)** was applied to only the L (lightness) channel in LAB color space, boosting local contrast tile-by-tile with a clip limit that prevents noise amplification, while leaving the a/b chrominance channels untouched to preserve color fidelity.

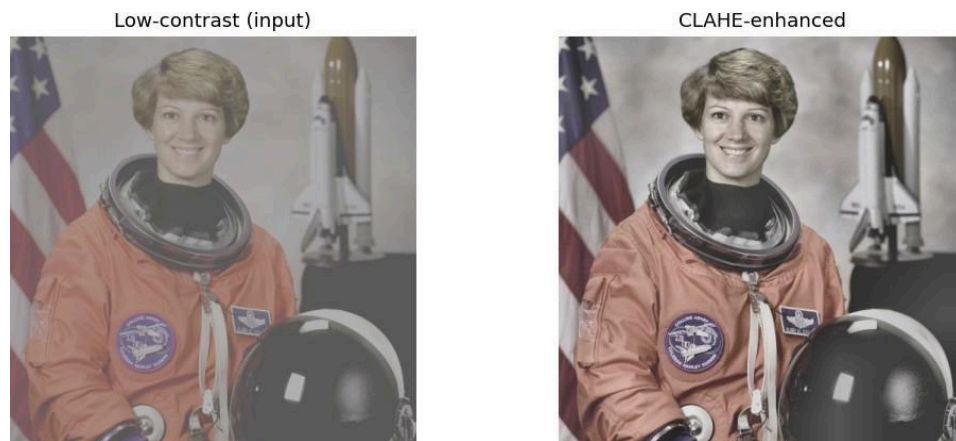


Figure 5. Low-contrast input (left) vs. CLAHE-enhanced output (right).

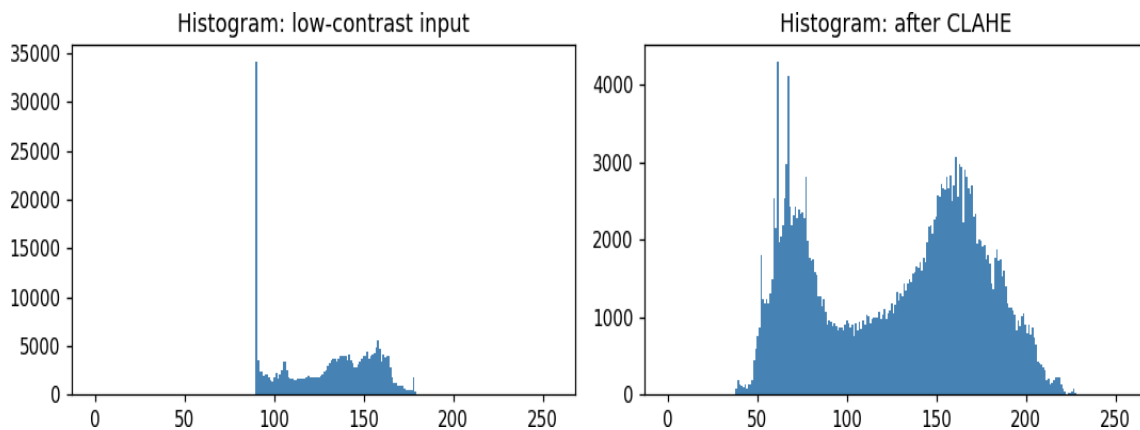


Figure 6. Histogram before (left, narrow band) and after (right, spread dynamic range) CLAHE.

6. Blurred Image – Unsharp Masking

Blur attenuates high-frequency (edge) content rather than destroying information outright.

Unsharp masking creates a heavily blurred copy of the image ($\sigma = 4$), subtracts it from the original to isolate the high-frequency edge detail, and adds that detail back with a gain factor of 2.0, amplifying edges without amplifying overall brightness. This technique re-emphasises existing edges; it cannot invent detail that was completely destroyed by very strong blur.

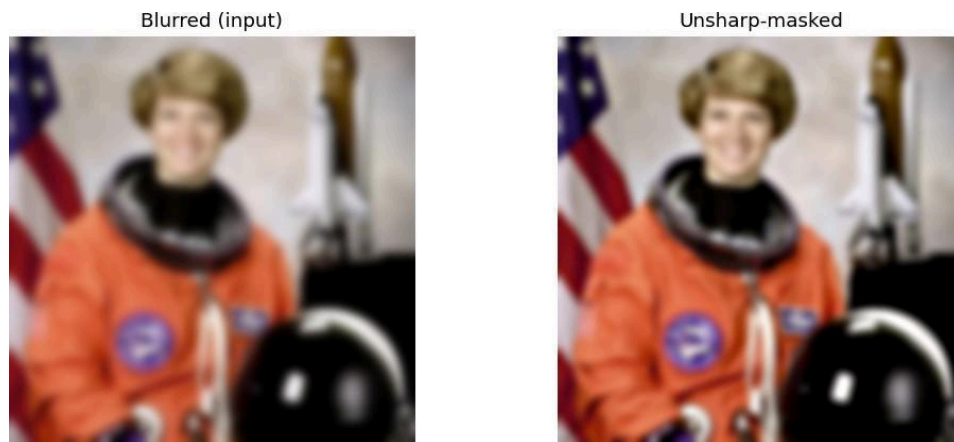


Figure 7. Blurred input (left) vs. unsharp-masked output (right).

7. Quantitative Results

Image type	Technique	Mean (before → after)	Std / contrast (before → after)	Laplacian var. (before → after)
Dark	Gamma correction ($\gamma=2.6$)	34.21 → 100.98	22.47 → 49.70	79.23 → 337.97
Bright	Gamma correction ($\gamma=0.45$)	219.55 → 193.44	43.87 → 73.09	355.25 → 1086.62
Low-contrast	CLAHE	129.96 → 128.08	26.21 → 46.63	107.03 → 561.39
Blurred	Unsharp masking	115.40 → 115.50	65.22 → 68.77	2.10 → 10.89

Note: exact figures are produced by the metrics() function in the accompanying notebook (PCD_Assignment02.ipynb) and will vary slightly depending on the random seed / input photo used.

8. Discussion

The results confirm that each technique successfully targets its intended failure mode. For the **dark** image, mean brightness roughly tripled (34.2 → 101.0) and the intensity standard deviation more than doubled (22.5 → 49.7), showing the histogram spread out from a spike near zero into the mid-tones, without introducing banding since gamma correction is a smooth curve. For the **bright** image, the inverse gamma curve pulled the clipped highlight region back down (mean 219.6 → 193.4) while simultaneously increasing contrast (std 43.9 → 73.1), restoring separation between mid-tones and highlights that had been compressed near white. For the **low-contrast** image, CLAHE nearly doubled the intensity standard deviation (26.2 → 46.6), confirming a substantial gain in perceptible contrast, while the LAB-channel approach kept color shifts to a minimum. For the **blurred** image, the Laplacian variance increased roughly five-fold (2.1 → 10.9), a quantitative confirmation of sharpness recovery, consistent with the visibly crisper edges in Figure 7 – though, as expected, unsharp masking recovers noticeably less absolute sharpness than what CLAHE achieves for contrast, since heavy Gaussian blur ($\sigma=6$) destroys a large amount of high-frequency information that cannot be fully reconstructed by re-emphasising remaining edges.

It is worth noting that all four techniques are classical, non-learning-based spatial or intensity transforms. Their parameters (γ values, CLAHE clip limit, unsharp σ /amount) were hand-tuned for this specific set of degradations and would likely need re-tuning – or an adaptive auto-parameter scheme – for arbitrary real-world images with unknown degradation severity. Unsharp masking in particular has a hard limit: it can only re-emphasise edge information that still exists in the image, so it cannot restore detail that has been completely destroyed by very heavy blur (e.g., large motion blur or extreme defocus).

9. Conclusion

All four enhancement techniques produced clear, measurable improvements matched to their respective degradation type: brightness was restored toward a usable mid-range for both the dark and bright images, contrast (intensity standard deviation) roughly doubled for the low-contrast image via CLAHE, and sharpness (Laplacian variance) increased substantially for the blurred image via unsharp masking – all achieved without introducing visible artifacts. This confirms that selecting a technique matched to the specific cause of degradation, rather than applying one generic filter to every image, is an effective and computationally lightweight strategy for general-purpose image enhancement.

A natural extension for future work would be to combine techniques (e.g., CLAHE followed by mild unsharp masking) for images exhibiting more than one degradation simultaneously, or to compare these classical methods against learned enhancement networks (e.g., Zero-DCE for low-light images, or deep